

A Literal Finite Boolean Algebra Obstruction to Question 5.8 of arXiv:2207.13990

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Source and Question

The source is Jerzy Kakol, Damian Sobota, and Lyubomyr Zdomskyy, *Grothendieck $C(K)$ -spaces and the Josefson–Nissenzweig theorem*, arXiv:2207.13990v2. Definition 5.6 says that a Boolean algebra $\mathcal{A}$ has the Grothendieck property, respectively the ℓ_1 -Grothendieck property, when its Stone space $\text{St}(\mathcal{A})$ has the corresponding property.

copy of c_0 but $C_p(K)$ does not contain any complemented copies of $(c_0)_p$.

We will also need the following definition.

Definition 5.6. A Boolean algebra $\mathcal{A}$ has the *Grothendieck property* (resp. the ℓ_1 -Grothendieck property) if its Stone space $\text{St}(\mathcal{A})$ has this property.

Recall that every (σ) -complete Boolean algebra has the Grothendieck property. By the Stone duality we also obtain the following corollary saying that every (σ) -complete Boolean algebra may be slimmed down in such a way that it loses the Grothendieck property but still preserves the ℓ_1 -Grothendieck property.

Corollary 5.7. *For every σ -complete Boolean algebra $\mathcal{A}$ there exists a subalgebra $\mathcal{B} \subseteq \mathcal{A}$ such that $\mathcal{B}$ does not have the Grothendieck property but it has the ℓ_1 -Grothendieck property.* $\square$

Analyzing the proof of Theorem 5.3, it seems that the theorem might also hold for those infinite totally disconnected compact spaces which are the Stone spaces of Boolean algebras with e.g. Haydon’s Subsequential Completeness Property ([24]). We do not know however how far the σ -completeness of $\mathcal{A}$ can be weakened in Corollary 5.7, which motivates the following question.

Question 5.8 asks whether every Boolean algebra with the Grothendieck property has a subalgebra which fails the Grothendieck property but still has the ℓ_1 -Grothendieck property.

Question 5.8. *Let $\mathcal{A}$ be a Boolean algebra with the Grothendieck property. Does there exist a Boolean subalgebra $\mathcal{B}$ of $\mathcal{A}$ which fails to have the Grothendieck property but which nonetheless has the ℓ_1 -Grothendieck property?*

6. SYSTEMS OF SIMPLE EXTENSIONS AND THE FSJNP

Idea of the Proof

The stated question omits an infinitude hypothesis. Finite Boolean algebras are immediate obstructions: their Stone spaces are finite, so the associated $C(K)$ -spaces are finite-dimensional. Finite-dimensional spaces have no room for a weak-star/weak sequential distinction, and every subalgebra remains finite.

Theorem 1. *Question 5.8 has a negative answer as literally stated.*

Proof. Let $\mathcal{A} = \{0, 1\}$ be the two-element Boolean algebra. Its Stone space $\text{St}(\mathcal{A})$ is a singleton. Hence $C(\text{St}(\mathcal{A}))$ is one-dimensional. Every finite-dimensional Banach space is Grothendieck: on a finite-dimensional dual space all Hausdorff locally convex vector topologies coincide, so any weak-star convergent sequence is weakly convergent. Thus $\mathcal{A}$ has the Grothendieck property in the sense of Definition 5.6.

The only Boolean subalgebra of $\mathcal{A}$ is $\mathcal{A}$ itself. Since $\mathcal{A}$ has the Grothendieck property, there is no subalgebra $\mathcal{B} \subseteq \mathcal{A}$ which fails to have the Grothendieck property. Therefore the existential conclusion in Question 5.8 is false for this $\mathcal{A}$. $\square$

Limitations

This is only a literal finite counterexample. It should be read as an erratum-style edge case: Question 5.8, as printed, does not include the hypothesis that $\mathcal{A}$ is infinite. The surrounding text and the preceding Corollary 5.7 strongly suggest that the intended problem is the infinite Boolean-algebra version.

This packet does not solve that intended infinite problem, and it does not improve the source's positive result for σ -complete Boolean algebras.

Follow-up on the Intended Infinite Version

After finding the finite obstruction, we made a separate serious attempt on the intended infinite version. No full infinite-case proof or counterexample was obtained.

The source proof for σ -complete algebras works by taking a zero-dimensional compact space $K = \text{St}(\mathcal{A})$, a non-atomic probability measure μ , and pairwise disjoint clopens A_n , then collapsing a closed set F determined by positive relative $\mu(A_n)$ -density. The quotient K/F is not Grothendieck. To prove that it is nevertheless ℓ_1 -Grothendieck, the source uses that the ideal

$$\mathcal{Z} = \left\{ C \subseteq K : C \text{ clopen and } \frac{\mu(A_n \cap C)}{\mu(A_n)} \rightarrow 0 \right\}$$

has a strong pseudo-union property. In the source, this property is supplied by basic disconnect-
edness, equivalently by σ -completeness of the clopen algebra.

The follow-up attempt indicates that the main obstruction to the infinite problem is precisely this pseudo-union step. Haydon’s Subsequential Completeness Property and related clopen-separation tools provide useful subsequence and approximate-separation statements, but they do not immediately force the required supremum to remain in the vanishing ideal $\mathcal{Z}$. Thus the most plausible direct route is to prove, or refute, that the Grothendieck property itself forces enough P -ideal behavior of $\mathcal{Z}$.

A final literature push sharpened this obstruction. Marciszewski–Sobota’s filter-space criterion suggests the following contrapositive route: if failure of the $\mathcal{Z}$ -pseudo-union property always produced a countable accumulating subspace $N_{\mathcal{F}}$ with the bounded Josefson–Nissenzweig property inside the original Grothendieck Stone space K , then K could not have been Grothendieck. The route did not close: pseudo-union failure gives positive relative-density limsups near F , while the filter criterion needs probability concentration in every neighborhood.

A small safe positive subcase remains available: if an infinite Grothendieck Boolean algebra contains an infinite σ -complete subalgebra, then the source Corollary 5.7 can be applied inside that subalgebra. The general infinite version remains open in this packet.

Novelty and Verification Notes

The run’s cheap indexes showed no existing packet for arXiv:2207.13990. A bounded web search for exact phrases around Question 5.8, including the phrases *Boolean algebra with the Grothendieck property*, `ell.1-Grothendieck`, and `subalgebra`, found the source paper and adjacent literature but no separate correction of this finite edge case. The argument itself uses only the source definition and the standard finite-dimensional observation above.

Human Review Recommendation

Review as an erratum-style literal counterexample. If Question 5.8 is read with the implicit hypothesis that $\mathcal{A}$ is infinite, this packet should be classified as a scope correction rather than as a solution of the intended problem. The accompanying run attempt notes record the unresolved infinite case and the current best obstruction.

Reference

Jerzy Kakol, Damian Sobota, Lyubomyr Zdomskyy, *Grothendieck $C(K)$ -spaces and the Josefson–Nissenzweig theorem*, arXiv:2207.13990v2, 2023.